Project_code

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The MLB doping era refers to a period from the 1980s to the late 2000s when widespread use of performance-enhancing drugs (PEDs) significantly inflated offensive statistics, notably home run records.

In the research paper by Glynn and Tokdar, they looked at MLB home-run data provided by Lahman's database from 1990-2014.

Now, we want to extend this study by looking at data beyond 2014. We will be looking at MLB data from 2010-2024 courtesy from Kaggle.

```r
data = read.csv("mlb_season_data.csv")
head(data)
```

```
##   season first_name last_name           link position team games_played at_bats
## 1   1876        Bob      Addy /player/110074        X  CHI           32     142
## 2   1876        Art   Allison /player/110170        X  LOU           31     130
## 3   1876       Doug   Allison /player/110172        C  HAR           44     163
## 4   1876       Fred    Andrus /player/110275        X  CHI            8      36
## 5   1876        Cap     Anson /player/110284       3B  CHI           66     309
## 6   1876       Ross    Barnes /player/110587       2B  CHI           66     322
##   runs hits doubles triples homeruns rbi walks strikeouts stolen_bases
## 1   36   40       4       1        0  16     5          0           --
## 2    9   27       2       1        0  10     2          6           --
## 3   19   43       4       0        0  15     3          9           --
## 4    6   11       3       0        0   2     0          5           --
## 5   63  110       9       7        2  59    12          8           --
## 6  126  138      21      14        1  59    20          8           --
##   caught_stealing batting_average on_base_percentage slugging_percentage
## 1              --           0.282                 --               0.324
## 2              --           0.208                 --               0.238
## 3              --           0.264                 --               0.288
## 4              --           0.306                 --               0.389
## 5              --           0.356                 --               0.450
## 6              --           0.429                 --               0.590
##   on_base_plus_slugging
## 1                    --
## 2                    --
## 3                    --
## 4                    --
## 5                    --
## 6                    --
```

```r
summary(data)
```

```
##      season      first_name          last_name             link          
##  Min.   :1876   Length:104067      Length:104067      Length:104067     
##  1st Qu.:1933   Class :character   Class :character   Class :character  
##  Median :1970   Mode  :character   Mode  :character   Mode  :character  
##  Mean   :1965                                                           
##  3rd Qu.:2001                                                           
##  Max.   :2024                                                           
##                                                                         
##    position             team            games_played       at_bats     
##  Length:104067      Length:104067      Min.   :  1.00   Min.   :  0.0  
##  Class :character   Class :character   1st Qu.: 15.00   1st Qu.:  8.0  
##  Mode  :character   Mode  :character   Median : 38.00   Median : 63.0  
##                                        Mean   : 55.38   Mean   :156.2  
##                                        3rd Qu.: 90.00   3rd Qu.:265.0  
##                                        Max.   :165.00   Max.   :716.0  
##                                                                        
##       runs             hits           doubles          triples     
##  Min.   :  0.00   Min.   :  0.00   Min.   : 0.000   Min.   : 0.000  
##  1st Qu.:  0.00   1st Qu.:  1.00   1st Qu.: 0.000   1st Qu.: 0.000  
##  Median :  6.00   Median : 12.00   Median : 2.000   Median : 0.000  
##  Mean   : 20.81   Mean   : 40.78   Mean   : 6.975   Mean   : 1.394  
##  3rd Qu.: 32.00   3rd Qu.: 67.00   3rd Qu.:11.000   3rd Qu.: 2.000  
##  Max.   :192.00   Max.   :262.00   Max.   :67.000   Max.   :36.000  
##                                                                     
##     homeruns          rbi                 walks          strikeouts   
##  Min.   : 0.000   Length:104067      Min.   :  0.00   Min.   :  0.0  
##  1st Qu.: 0.000   Class :character   1st Qu.:  0.00   1st Qu.:  2.0  
##  Median : 0.000   Mode  :character   Median :  4.00   Median : 12.0  
##  Mean   : 3.255                      Mean   : 14.47   Mean   : 24.5  
##  3rd Qu.: 3.000                      3rd Qu.: 22.00   3rd Qu.: 35.0  
##  Max.   :73.000                      Max.   :232.00   Max.   :223.0  
##                                                       NA's   :11966  
##  stolen_bases       caught_stealing    batting_average on_base_percentage
##  Length:104067      Length:104067      Min.   :0.000   Length:104067     
##  Class :character   Class :character   1st Qu.:0.105   Class :character  
##  Mode  :character   Mode  :character   Median :0.222   Mode  :character  
##                                        Mean   :0.192                     
##                                        3rd Qu.:0.270                     
##                                        Max.   :1.000                     
##                                                                          
##  slugging_percentage on_base_plus_slugging
##  Min.   :0.0000      Length:104067        
##  1st Qu.:0.1210      Class :character     
##  Median :0.2940      Mode  :character     
##  Mean   :0.2686                           
##  3rd Qu.:0.3920                           
##  Max.   :4.0000                           
## 
```

```r
data = data %>%
  mutate(across(c(first_name, last_name), ~stri_trans_general(., "latin-ascii"))) %>%
  mutate(name = paste(first_name, last_name), .after = season) %>%
  mutate(player_id = str_extract(link, "\\d+"), .before = name) %>%
  mutate(player_id = as.numeric(player_id)) %>%
  filter(season >= 2010) %>%
  select(!c(first_name, last_name, link))
```

We don't have the age of each player in which was used to model the expected aging curve of a player's home run production. Therefore, we need to first find a way to add the player's age for that season using another dataset and the baseballr package.

```r
chadwick = chadwick_player_lu()
id_map = chadwick %>%
  select(key_mlbam, birth_year) %>%
  filter(!is.na(key_mlbam))

data = data %>%
  left_join(id_map, by = c("player_id" = "key_mlbam")) %>%
  mutate(age = season - birth_year)
```

Data Summary

```r
summary(data)
```

```
##      season       player_id          name             position        
##  Min.   :2010   Min.   :110029   Length:16514       Length:16514      
##  1st Qu.:2013   1st Qu.:456501   Class :character   Class :character  
##  Median :2016   Median :541642   Mode  :character   Mode  :character  
##  Mean   :2016   Mean   :527195                                        
##  3rd Qu.:2020   3rd Qu.:607208                                        
##  Max.   :2024   Max.   :808982                                        
##      team            games_played       at_bats            runs       
##  Length:16514       Min.   :  1.00   Min.   :  0.0   Min.   :  0.00  
##  Class :character   1st Qu.: 19.00   1st Qu.:  1.0   1st Qu.:  0.00  
##  Mode  :character   Median : 42.00   Median : 42.0   Median :  3.00  
##                     Mean   : 57.05   Mean   :143.5   Mean   : 18.66  
##                     3rd Qu.: 86.00   3rd Qu.:244.0   3rd Qu.: 30.00  
##                     Max.   :162.00   Max.   :684.0   Max.   :149.00  
##       hits        doubles          triples          homeruns     
##  Min.   :  0   Min.   : 0.000   Min.   : 0.0000   Min.   : 0.000  
##  1st Qu.:  0   1st Qu.: 0.000   1st Qu.: 0.0000   1st Qu.: 0.000  
##  Median :  7   Median : 1.000   Median : 0.0000   Median : 0.000  
##  Mean   : 36   Mean   : 7.134   Mean   : 0.6973   Mean   : 4.645  
##  3rd Qu.: 59   3rd Qu.:12.000   3rd Qu.: 1.0000   3rd Qu.: 6.000  
##  Max.   :225   Max.   :59.000   Max.   :16.0000   Max.   :62.000  
##      rbi                walks          strikeouts  stolen_bases      
##  Length:16514       Min.   :  0.00   Min.   :  0   Length:16514      
##  Class :character   1st Qu.:  0.00   1st Qu.:  1   Class :character  
##  Mode  :character   Median :  2.00   Median : 13   Mode  :character  
```

```
##                       Mean   : 13.21   Mean   : 34
##                       3rd Qu.: 20.00   3rd Qu.: 57
##                       Max.   :145.00   Max.   :222
##  caught_stealing      batting_average  on_base_percentage slugging_percentage
##  Length:16514         Min.   :0.0000   Length:16514       Min.   :0.0000
##  Class :character     1st Qu.:0.0000   Class :character   1st Qu.:0.0000
##  Mode  :character     Median :0.1930   Mode  :character   Median :0.2760
##                       Mean   :0.1548                      Mean   :0.2385
##                       3rd Qu.:0.2520                      3rd Qu.:0.4000
##                       Max.   :1.0000                      Max.   :4.0000
##  on_base_plus_slugging   birth_year        age
##  Length:16514          Min.   :1962   Min.   :19.00
##  Class :character      1st Qu.:1984   1st Qu.:26.00
##  Mode  :character      Median :1988   Median :28.00
##                        Mean   :1988   Mean   :28.54
##                        3rd Qu.:1992   3rd Qu.:31.00
##                        Max.   :2004   Max.   :50.00
```

```r
nrows = nrow(data)
nplayers = length(unique(data$player_id))
year_range = range(data$season, na.rm = TRUE)

summary_tbl = data.frame(
  Quantity = c("Player-seasons", "Unique players", "Season range",
               "Mean HR", "Median HR", "Max HR"))

Value = c(
    format(nrows, big.mark = ","),
    format(nplayers, big.mark = ","),
    paste(year_range, collapse = "-"),
    sprintf("%.2f", mean(data$homeruns, na.rm = TRUE)),
    sprintf("%.0f",  median(data$homeruns, na.rm = TRUE)),
    sprintf("%.0f",  max(data$homeruns, na.rm = TRUE))
  )
summary_tbl = summary_tbl %>%
  mutate(" " = Value)
kable(summary_tbl, caption = "Dataset overview.")
```

Table 1: Dataset overview.

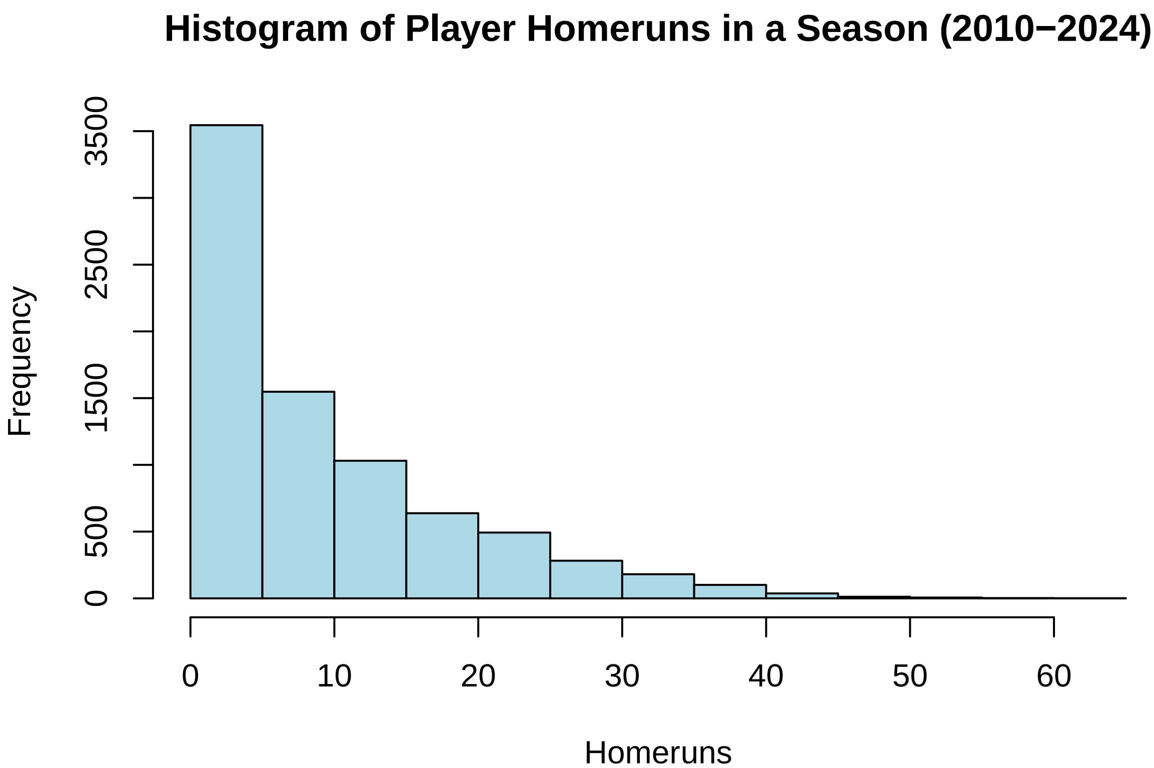
Quantity	
Player-seasons	16,514
Unique players	4,222
Season range	2010-2024
Mean HR	4.65
Median HR	0
Max HR	62

This is every player. Recall that most players only pitch or not/rarely bat at all. Some players could also be off the bench players who did not play much, so there is not that much data behind them. Recall that an average starting in the MLB gets about 500-700 plate appearances in an MLB season. Therefore we should only look at players who bat at least 50 times in a season and analyze their home-run data from that.

```r
data_batters = data %>%
  filter(at_bats > 50)
```

Now let's plot a histogram distribution of player homeruns in a season from 2010-2024.

```r
hist(data_batters$homeruns,
     breaks = 20,
     main = "Histogram of Player Homeruns in a Season (2010-2024)",
     xlab = "Homeruns",
     col = "lightblue"
     )
```



Let's see the data summaries again for only players with 50+ plate appearances in a season from 2010-2024.

```r
nrows = nrow(data_batters)
nplayers = length(unique(data_batters$player_id))
year_range = range(data_batters$season, na.rm = TRUE)

summary_tbl = data.frame(
  Quantity = c("Player-seasons", "Unique players", "Season range",
               "Mean HR", "Median HR", "Max HR"))

Value = c(
    format(nrows, big.mark = ","),
    format(nplayers, big.mark = ","),
    paste(year_range, collapse = "-"),
```

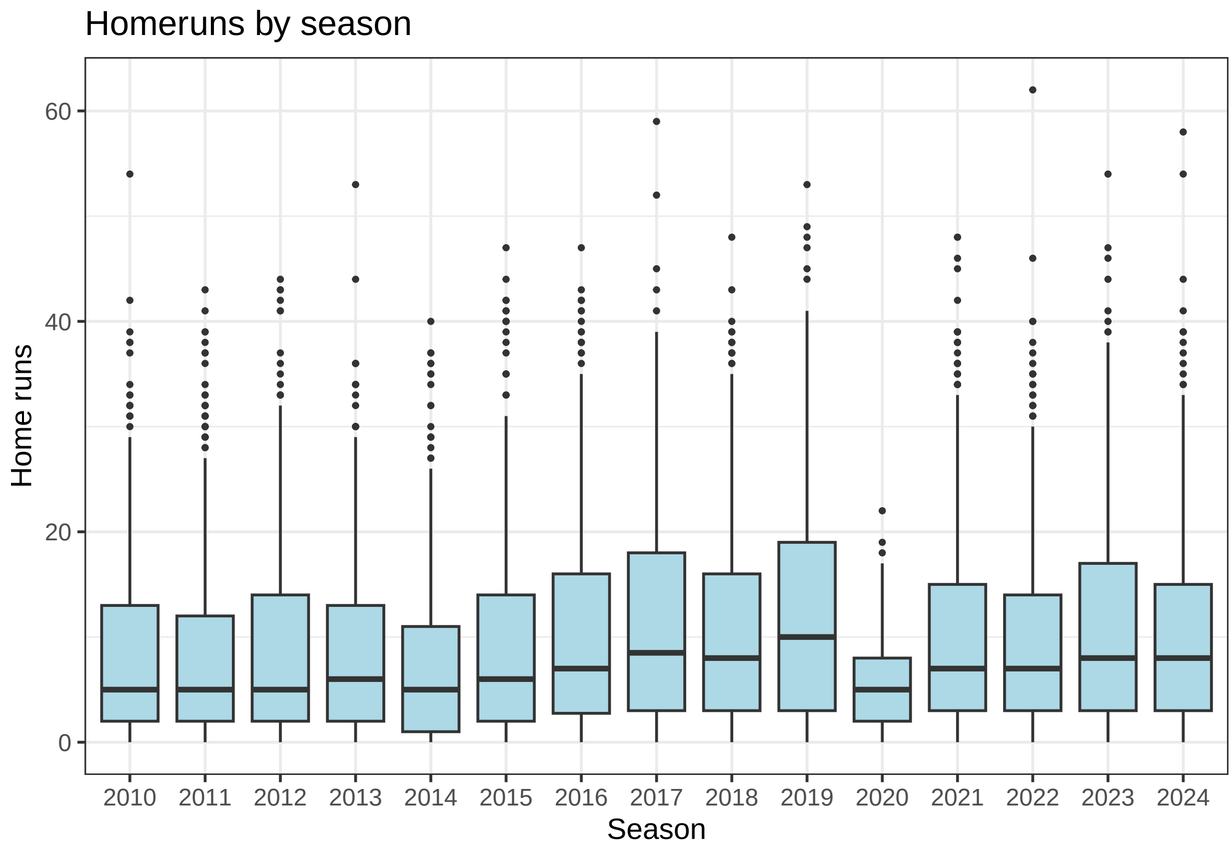
```r
    sprintf("%.2f", mean(data_batters$homeruns, na.rm = TRUE)),
    sprintf("%.0f",  median(data_batters$homeruns, na.rm = TRUE)),
    sprintf("%.0f",  max(data_batters$homeruns, na.rm = TRUE))
  )
summary_tbl = summary_tbl %>%
  mutate(" " = Value)
kable(summary_tbl, caption = "Dataset overview.")
```

Table 2: Dataset overview.

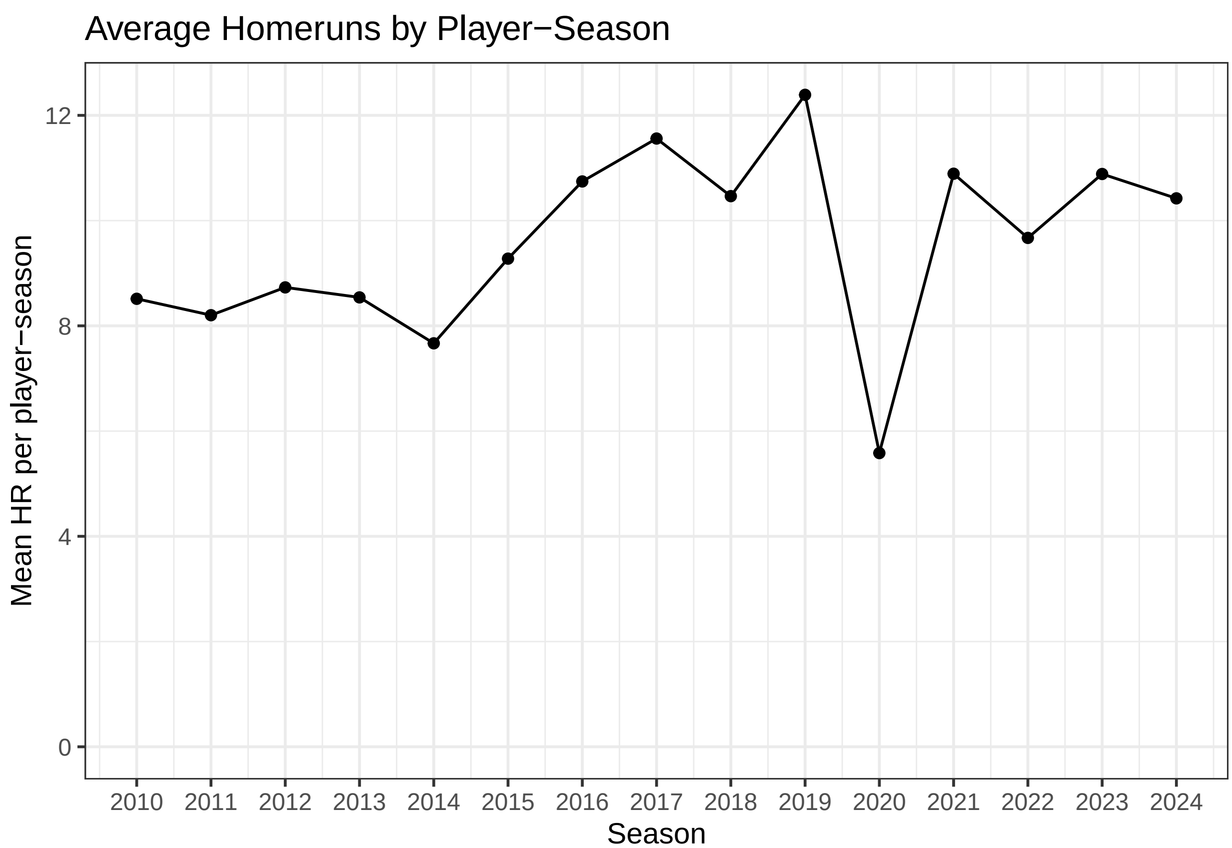
Quantity	
Player-seasons	7,877
Unique players	1,887
Season range	2010-2024
Mean HR	9.63
Median HR	7
Max HR	62

Exploratory Analysis

```r
ggplot(data_batters, aes(x = factor(season), y = homeruns)) +
  geom_boxplot(outlier.size = 0.6, fill = "lightblue") +
  labs(title = "Homeruns by season", x = "Season", y = "Home runs") +
  theme_bw()
```




```
data_batters %>%
  group_by(season) %>%
  summarise(mean_HR = mean(homeruns, na.rm = TRUE),
            mean_AB = mean(at_bats, na.rm = TRUE), .groups = "drop") %>%
  ggplot(aes(season, mean_HR)) +
  geom_line() +
  geom_point() +
  coord_cartesian(ylim = c(0, NA)) +
  scale_x_continuous(breaks = year_range[1]:year_range[2]) +
  labs(title = "Average Homeruns by Player-Season",
       x = "Season",
       y = "Mean HR per player-season") +
  theme_bw()
```



There is an increasing trend in average home runs per player-season from 2014 to 2019.

Recall that there was a historic home run surge from 2014 to 2019 was driven by a major alteration in the manufacturing of the baseball (reducing aerodynamic drag) and an analytical shift in player hitting styles toward higher launch angles.

Notice that in 2020, there is a big drop in average home runs. The reason for this is that 2020 was the COVID season where only roughly 60 games were played instead of the typical 162 game season.

A key concern with when trying to identify abnormal home run production relative to a baseline expectation, but the baseline level of home run production may vary substantially across seasons due to league-wide factors. For example, changes in the baseball, rule changes, and shortened seasons can shift overall home run rates in ways that are unrelated to an individual player's underlying ability or performance. If these season-level effects are not accounted for, the model may incorrectly classify players as abnormal simply because they played in a season with unusually high league-wide home run production. In particular, hitters in a season such as 2019 may appear abnormal not only because of their individual performance, but also because the entire league experienced elevated home run totals. Therefore, league wide HR changes (ball composition, rule changes) might be confounders for individual abnormality detection.

```
top_hr = data_batters %>%
  arrange(desc(homeruns)) %>%
  select(season, name, team, age, at_bats, homeruns,
         batting_average, on_base_plus_slugging) %>%
  head(10)
kable(top_hr, caption = "Top 10 single-season HR totals in the sample.",
      digits = 3,
      col.names = c("Season", "Name", "Team", "Age",
                    "At Bats", "Homeruns", "Average", "OPS"))
```

Table 3: Top 10 single-season HR totals in the sample.

Season	Name	Team	Age	At Bats	Homeruns	Average	OPS
2022	Aaron Judge	NYY	30	570	62	0.311	1.111
2017	Giancarlo Stanton	MIA	28	597	59	0.281	1.007
2024	Aaron Judge	NYY	32	559	58	0.322	1.159
2010	Jose Bautista	TOR	30	569	54	0.260	.995
2023	Matt Olson	ATL	29	608	54	0.283	.993
2024	Shohei Ohtani	LAD	30	636	54	0.310	1.036
2013	Chris Davis	BAL	27	584	53	0.286	1.004
2019	Pete Alonso	NYM	25	597	53	0.260	.941
2017	Aaron Judge	NYY	25	542	52	0.284	1.049
2019	Eugenio Suarez	CIN	28	575	49	0.271	.930